

MLE Sheet 9 Task 1.

1.1. Proof: Easy to show as

$$k_1(x, x') \geq 0 \wedge k_2(x, x') \geq 0$$

$$\Rightarrow k_1(x, x') + k_2(x, x') \geq 0. \text{ Thus } k_{\text{sum}} \text{ psd } \square.$$

1.2. (a) For the GP, we have the assumption that: (K_θ is the approximation to covariance matrix)
 $f(x) \sim N(0, K_\theta \text{ ~~(x, y)~~})$, ($m(x) = 0$)
 From the task, $\varepsilon \sim N(0, \sigma^2 I_n)$ was given.

Hence, $y|x \sim f(x) + \varepsilon \sim N(0, K_\theta \text{ ~~(x, y)~~ } + \sigma^2 I_n)$.

$$p(y|x, \theta) = \frac{1}{\sqrt{(2\pi)^n |K_\theta \text{ ~~(x, y)~~ } + \sigma^2 I_n|}} \exp\left(-\frac{1}{2} y^T (K_\theta \text{ ~~(x, y)~~ } + \sigma^2 I_n)^{-1} y\right)$$

which gives the log-likelihood as stated:

$$\log p(y|x, \theta) = -\frac{1}{2} y^T (K_\theta + \sigma^2 I_n)^{-1} y - \frac{1}{2} \log |K_\theta + \sigma^2 I_n| - \frac{n}{2} \log(2\pi)$$

(b) Data Fit: $-\frac{1}{2} y^T (K_\theta + \sigma^2 I_n)^{-1} y$.

The better K_θ approximates the real covariance matrix, the ~~better~~ closer to 0 is this term for the training data.

Complexity Penalty: $-\frac{1}{2} \log |K_\theta + \sigma^2 I_n|$.

If K_θ is too complicated (too many nonzero entries; entries too large), this penalty term will punish the complexity by making the log-likelihood smaller.

Why Occam's razor?: Occam's Razor encourages simple models too avoid overfitting, which agrees with the second term.